

# Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

## Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

## Classified Practice

## Differentiation and Tangent Lines

Q001 Written

**Differentiate the four Warm Up demands:**  $y = x^2 - 3x + 2$ ,  $y = (2x + 1)^3$ ,  $V = \pi r^2 h$  with respect to (r), and find  $f'(-1)$  for  $f(x) = x^3 - 3x + 1$ .

**Q002** Written

**Differentiate the six printed Try functions, the two variable-labelled functions  $S = \pi r^2$  and  $h = vt - \frac{1}{2}gt^2$ , and evaluate  $f'(2)$ ,  $f'(-3)$  for  $f(x) = 4x^3 - 3x^2 + 2$ .**

## Classified Practice

## Differentiation and Tangent Lines

Q003 **Written**

**Differentiate every given Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.**

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## Classified Practice

## Differentiation and Tangent Lines

Q004 Written

For  $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$ , find  $f'(1) + f'(-1)$ .

**Q005** Written

**Find the quadratic  $f(x) = ax^2 + bx + c$  satisfying  $f'(0) = -2$ ,  $f'(2) = 2$ ,  $f(1) = 2$ .**

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## Classified Practice

## Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying  $f'(0) = 2$ ,  $f'(1) = 4$ ,  $f(2) = 6$ .

**Q007** Written

**Find the four quadratics satisfying the four given triples of derivative and function conditions.**

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## Classified Practice

## Differentiation and Tangent Lines

Q008 Written

Find the quadratic  $f(x)$  satisfying  $f'(2) + f'(4) = 0$ ,  $f(3) = 0$ ,  $f(1) = 12$ .

**Q009** Written**Find the tangent line to  $y = x^2 - 4x + 1$  at  $(3, -2)$ .**

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## Classified Practice

## Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three given functions at their stated points:  $y = x^2 + 3$  at  $(-2, 7)$ ,  $y = x^2 - 3x + 2$  at  $((1, 0))$ , and  $y = -x^3 + 2x + 4$  at  $(-2, 8)$ .

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**Q011** Written**Find the tangent line for all nine given Exercise functions at their given points.**

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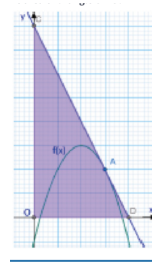
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## Classified Practice

## Differentiation and Tangent Lines

Q012 Written

For  $f(x) = -x^2 + 4x - 1$ , the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



## Classified Practice

## Differentiation and Tangent Lines

Q013 Written

Solve the two given Warm Up demands: tangents from  $(C(0,2))$  to  $y = x^2 - 3x + 6$ , and tangents to  $y = x^3 + 4x^2 - 3$  with gradient 3.

**Q014** Written

**Solve the two given Try demands: tangents from  $C(1, -1)$  to  $y = x^2 - x + 3$ , and tangents to  $y = x^3 + 2x - 3$  with gradient 5.**

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## Classified Practice

## Differentiation and Tangent Lines

Q015 **Written****Solve all six given** Exercise demands involving external points or prescribed gradients.

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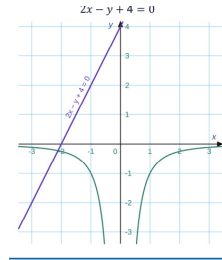


## Classified Practice

## Differentiation and Tangent Lines

Q016 Written

For  $f(x) = -1/x^2$ , find the tangent line parallel to  $2x - y + 4 = 0$ .



## Classified Practice

## Differentiation and Tangent Lines

Q017 MCQ

**The derivative of  $x^n$  is:****Choose the correct option.**

A  $x^{n-1}$

B  $nx^{n-1}$

C  $n + x$

D  $x^n$

Q018 MCQ

The derivative of  $V = \pi r^2 h$  with respect to  $r$  is:

Choose the correct option.

A  $\pi h$

B  $2\pi r h$

C  $2\pi r$

D  $\pi r^2$

## Classified Practice

## Differentiation and Tangent Lines

Q019 MCQ

**For  $f(x) = x^3 - 3x + 1$ ,  $f'(-1)$  is:****Choose the correct option.**

A -6

B -3

C 0

D 3

**Q020 MCQ****The challenge function can be written as:****Choose the correct option.**

**A**  $x^{5/3} + x^{-2}$

**B**  $x^5 + x^2$

**C**  $x^{3/5} + x^2$

**D**  $x^{5/3} + x^2$

## Classified Practice

## Differentiation and Tangent Lines

Q021 MCQ

For  $f(x) = ax^2 + bx + c$ ,  $(f'(x))$  is:

Choose the correct option.

A  $ax + b$

B  $2ax + b$

C  $2a + b + c$

D  $ax^2 + b$

**Q022 MCQ****The condition  $f'(0) = -2$  gives:****Choose the correct option.**

**A**  $a = -2$

**B**  $b = -2$

**C**  $c = -2$

**D**  $2a = -2$

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## Classified Practice

## Differentiation and Tangent Lines

Q023 MCQ

**The warm-up quadratic is:****Choose the correct option.**

A  $x^2 - 2x + 3$

B  $x^2 + 2x + 3$

C  $2x^2 - x + 3$

D  $x^2 - 2x - 3$



Q024 MCQ

**The challenge coefficient  $a$  is:****Choose the correct option.**

A 1

B 2

C 3

D 6

## Classified Practice

## Differentiation and Tangent Lines

Q025 MCQ

**The tangent gradient at  $x=a$  is:****Choose the correct option.**A  $f(a)$ B  $a$ C  $f'(a)$ D  $1/f'(a)$

Q026 MCQ

**The tangent to  $y = x^2 - 4x + 1$  at  $(3, -2)$  is:****Choose the correct option.**

A  $y = x - 5$

B  $y = 2x - 8$

C  $y = -2x + 4$

D  $y = 3x - 11$

## Classified Practice

## Differentiation and Tangent Lines

Q027 MCQ

**In the triangle challenge, the x-intercept of the tangent is:****Choose the correct option.**

A 2

B 3

C 4

D 8

**Q028** MCQ**The area of triangle (ODC) in the challenge is:****Choose the correct option.****A** 8**B** 12**C** 16**D** 32

## Classified Practice

## Differentiation and Tangent Lines

Q029 MCQ

**When the contact point is not given, set it to:****Choose the correct option.**A  $((0,0))$ B  $((a,f(a)))$ C  $((f(a),a))$ D  $((1,f'(a)))$

**Q030** MCQ**The gradient of  $2x - y + 4 = 0$  is:****Choose the correct option.****A** -2**B** -1**C** 1**D** 2

## Classified Practice

## Differentiation and Tangent Lines

Q031 MCQ

For  $f(x) = -1/x^2$ ,  $(f'(x))$  is:

Choose the correct option.

A  $-2/x^3$

B  $2/x^3$

C  $1/x^2$

D  $-1/x^3$



**Q032** MCQ**The parallel tangent in the challenge is:****Choose the correct option.**

**A**  $y = 2x + 3$

**B**  $y = 2x - 3$

**C**  $y = -2x - 3$

**D**  $y = x - 2$

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## Classified Practice

## Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written

**Differentiate**  $3x^3 - 2x$ .

**Quiz WRITTEN 2** Concept Quiz · Written

**Find the quadratic with  $f'(0) = 1, f'(1) = 3, f(1) = 2$ .**

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## Classified Practice

## Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to  $y = x^2$  at  $((2,4))$ .

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**Quiz WRITTEN 4** Concept Quiz · Written

**For  $f(x) = -1/x$ , find the tangent parallel to  $y = x + 2$ .**

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## Classified Practice

## Differentiation and Tangent Lines

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

**The derivative of a constant is:**

**Choose the correct option.**

A 0

B 1

C x

D the constant

**Quiz MCQ 6** Concept Quiz · MCQ

**For  $f(x) = ax^2 + bx + c$ , the coefficient equations come from:**

**Choose the correct option.**

- A three function values
- B two derivatives and one function value
- C one derivative only
- D the tangent gradient only

## Classified Practice

## Differentiation and Tangent Lines

Quiz MCQ 7 **Concept Quiz · MCQ****The tangent equation in point-slope form is:****Choose the correct option.**

A  $y - y_1 = m(x - x_1)$

B  $y = mx + c$

C  $y + y_1 = m(x + x_1)$

D  $x - x_1 = m(y - y_1)$



**Quiz MCQ 8** Concept Quiz · MCQ

**A tangent parallel to  $y = 3x - 1$  has gradient:**

**Choose the correct option.**

A -3

B  $1/3$

C 3

D 1

# Answer key

Use the question number shown on each page.

**Q001**  $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

**Q008**  $f(x) = 3x^2 - 18x + 27$

**Q019** C

**Q002**  $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

**Q009**  $y = 2x - 8$

**Q020** A

**Q003** Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4  $-24, -3$ ; Exercise 5 (3,28).

**Q010**  $y = -4x - 1; y = -x + 1; y = -10x - 12$

**Q021** B

**Q011**  $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

**Q022** B

**Q012** 16 square units

**Q023** A

**Q013**  $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

**Q024** C

**Q014**  $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

**Q025** C

**Q004**  $\frac{10}{3}$

**Q015**  $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

**Q026** B

**Q005**  $f(x) = x^2 - 2x + 3$

**Q016**  $y = 2x - 3$

**Q027** C

**Q006**  $f(x) = x^2 + 2x$

**Q007**  $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

**Q017** B

**Q028** C

**Q018** B

**Q029** B

**Q030** D

**Q031** B

**Q032** B

**Quiz WRITTEN 1**  $9x^2 - 2$

**Quiz WRITTEN 2**  $x^2 + x$

**Quiz WRITTEN 3**  $y = 4x - 4$

**Quiz WRITTEN 4**  $y = x - 4$

**Quiz MCQ 5** A

**Quiz MCQ 6** B

**Quiz MCQ 7** A

**Quiz MCQ 8** C